

YASH JOSHI

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SUMMARY

Senior React/Next.js engineer with 5+ years building high-performance, scalable applications across AI simulation, fintech, healthcare, and automotive domains. Recently shifted into AI-evaluation engineering — authoring containerized benchmark tasks with hybrid verifiers and calibrated rubrics that measure how well frontier coding agents perform on realistic software-engineering work — including a React-virtualization benchmark that reliably defeats both Opus 4.7 and GPT-5-Codex. Comfortable across the stack: frontend architecture (Next.js App Router, Zustand, Recharts), Node/PHP services, PostgreSQL data modeling, and Docker-based evaluation infrastructure, with strong instincts for observability, type safety, and production-grade delivery.

SKILLS

Frontend: React.js, Next.js (App Router, SSR/ISR, Server Actions), TypeScript, JavaScript (ES6+), Tailwind CSS, Chakra UI, Headless UI v2, Radix UI, Zustand, nuqs, Recharts, d3, Framer Motion, React Hook Form, Formik, FullCalendar, Lottie React, react-pdf, react-share, html2canvas-pro, Razorpay, list virtualization

Backend & Data: Node.js, PHP, PostgreSQL, SQL query optimization (N+1, indexing, caching), REST & AJAX APIs, database migrations & seeding, Pino structured logging, JWT/OTP auth

AI Evaluation: Harbor benchmark task authoring, golden/oracle solutions, interactive LLM rubric judges, deterministic performance gates, headless-Chromium/Playwright behavioral measurement, model calibration (Opus 4.7, GPT-5-Codex) & de-leaking, Docker eval environments, opencode harness, Python, Bash

Observability: OpenTelemetry, Grafana Faro, distributed tracing, sourcemap management, pg_stat_statements

Testing & Quality: Playwright (multi-browser E2E), ESLint, Prettier, Husky, Commitlint, Conventional Commits

Practices & Tools: Git, Agile, Documentation, Device Fingerprinting, Consent Management, pnpm, Turbopack

WORK EXPERIENCE

Senior Software Engineer — Theta Software

Dec 2025 – June 2026

AI Simulation Platform that translates real-world workspaces into simulation environments to train expert-level agents.

Theta — AI-Evaluation Benchmark Engineering

Harbor-format benchmark system that evaluates AI coding agents on realistic, multi-phase software-engineering tasks.

- Authored a **React-virtualization benchmark that defeats frontier coding agents**: the reference (oracle) solution scores **0.96** while **Opus 4.7 (0.36)** and **GPT-5-Codex (0.345)** both fail it across k=3 trials (no-op floor 0.03) — a robust discriminator across model families, where prior backend-perf tasks couldn't push any model below 0.5.
- Authored end-to-end benchmark tasks across **three real codebases** — Excalidraw feature extensions (a Frames-Navigator sidebar with fractional-index reordering; a cancellable async image-upload pipeline with undo/redo, AbortController cancellation, and accessible in-flight announcements), OpenEMR backend-performance (N+1 elimination, reference-data caching, a clinical-reminder engine), and the Ramp virtualization task above — each with task spec, golden solution, verifier, and scoring rubric.
- Engineered a hybrid verifier in Python combining deterministic performance gates (pg_stat_statements query counts, golden-response diffing, headless-Chromium/Playwright behavioral measurement) with an interactive LLM rubric judge and a behavioral gate, for composite, defensible, hard-to-game scoring.
- Built reproducible single-container Docker evaluation environments (multi-stage build, PostgreSQL 16 + PHP/Apache, seeded with ~500 patients and 1,550 insurance rows) and replaced a bind-mount design that was leaking oracle solutions across runs — a contamination bug that had been inflating benchmark scores.
- Established through four calibrated negative results that self-verifiable backend-perf tasks are solved by frontier models and can't discriminate them — pivoting to a class they can't self-check; de-leaked task specifications (stripping gate names, thresholds, and solution hints) to prevent agent-run contamination.
- Validated oracle solutions through iterative calibration cycles (encounter queries 64→0, chart 10→3) and hardened the harness against edge cases: CVE-2022-24765 git-ownership fix, rsync symlink dangles, and zero-query cache-hit gates.

Theta EMR — AI Training Environment for Clinical Workflows

OpenEMR customized into a training platform where autonomous agents learn clinical and billing workflows against golden references.

- Built an interactive growth-charts system covering CDC, WHO, Fenton Down Syndrome, and Pre-Term standards (30+ percentile datasets) with crosshair navigation, hover info panels, and dynamic chart selection inside the appointment dialog.
- Redesigned the clinical calendar (day/week/month views) with click-to-create scheduling, exam-room filtering, color-coded appointments, and an appointment-preview hover sidebar surfacing patient demographics across all views.
- Delivered a multi-tab Schedule Appointment dialog integrating patient creation, break management, MU Helper maternal-child demographics, consent-form multi-select, and growth charts — with tab-switch validation to prevent unsaved-data loss.
- Architected an Advanced Report engine: ~82 backend SQL filter builders across Appointments/Patients/Procedures tabs, multi-status queries, save/load/delete saved filters, datepicker filtering, and export.
- Shipped a billing suite — Live Claims Feed (real-time status pipeline, ar_activity financial columns, schedule-service modal), Billing Summary (income analysis, payment-speed SLAs), and Adjustment Master CRUD.
- Built provider settings pages — eRx Settings (3 tabs + pharmacy/favorites REST API), API Management (API + ConnectEHR), and Appointment/Billing Profiles & Templates — each as a DB migration + PHP page + REST/AJAX API + CSS/JS.
- Added a Recurring Events management page (stop/end-date workflows, doctor/break filters, soft-delete audit columns), Manage Macro Buttons CRUD for clinical shortcuts, and Invite-OnPatient letter flow.

Theta Dynamics — Dynamics 365 CRM Replica

Pixel-perfect Microsoft Dynamics 365 sales-pipeline clone in Next.js 14, TypeScript, Tailwind, and Zustand.

- Architected the CRM clone following Microsoft Fluent Design — server/client component split, config extraction, portal-based rendering, and a design-token system — replicating the full sales-pipeline experience.

- Built an advanced Recharts visualization suite: Deal Tracker bubble chart (date/probability/revenue axes), click-to-filter horizontal bar charts, sales funnel, and territory progress charts, with chart-to-table filtering driven in real time.
- Designed the PostgreSQL data layer — accounts, contacts, opportunities, and forecast tables with seed data — and migrated user/lead primary keys from BIGINT to UUID across all foreign keys for referential integrity.
- Converted the forecasts module to DB-backed data via Next.js Server Actions (periods, forecasts, drill-down records, adjustments), adding an adjustment dialog with audit history and an inline-editable underlying-records drill-down panel.
- Implemented Zustand slice-pattern state for leads, opportunities, and tasks — multi-filter, full-text search, multi-select operations — plus a tasks kanban with focused view, inline edit, and data import.
- Migrated the component library to Headless UI v2 and added Radix UI tooltips for accessibility; fixed portal-dropdown overflow clipping and drill-down cache invalidation on customer changes.

Senior Software Engineer — Automint

Jan 2025 – Dec 2025

Product: Icemint — a device-leasing (Device-as-a-Service) platform where employees lease phones, laptops, and cars through their employer, with eligibility rules, tax savings, foreclosure, orders, and payments. Built in Next.js (App Router), TypeScript, Chakra UI, and Zustand.

- Authored and published **device-unique-keygen**, an open-source npm package for browser device fingerprinting, and integrated it to power guest authentication and anonymous tracking — enabling sessionless user visibility without login.
- Built the lease tax-savings calculator: lease-vs-loan comparison with real-time monthly rental, old/new tax-regime deductions, and eligibility thresholds across 36–48 month tenures, using custom React hooks, Zustand, and typed tenure models.
- Developed the multi-step order workflow (employee details → address → Razorpay payment) as a state machine with payment-redirect handling, session persistence, and multiple-order support; built the device & accessories catalog with bundle pricing and a foreclosure (early-exit) calculator showing month-by-month amortization, residual value, and fees.
- Implemented the onboarding flow (email/mobile OTP → salary/CTC → device selection) with enum-based step routing and input validation, and an eligibility-rules engine gating quotes on monthly-rental caps and CTC thresholds; added a request-callback feature and consent management with config-driven toggles.
- Implemented end-to-end observability with OpenTelemetry and Grafana Faro — W3C traceparent propagation through Axios interceptors and Pino server logs enriched with trace/span IDs — plus a singleton FaroManager with an email/deviceId session strategy and sourcemap upload to Grafana for de-minified error attribution.
- Refactored authentication from a brittle isLoggedIn flag to a single-source-of-truth user store (Authorization cookie + UserInitializer), and removed redundant user API calls across profile/company/personal pages to cut network traffic and stale-state bugs.
- Built a modular Next.js middleware stack (route protection, CORS, sourcemap access control, bot detection, onboarding redirects) and architected Zustand state with cookie/localStorage sync and nuqs URL state for shareable deep links.
- Set up multi-browser Playwright E2E over HTTPS (Chromium, Firefox, WebKit) with CI retries, structured Pino logging for request correlation, and dual JWT/OTP auth for guest and registered users.
- Reduced debugging time ~60% via distributed tracing and sourcemaps; cut local build times ~40% with Turbopack; improved mobile load times ~35% with PWA offline support.

Software Engineer — BigBinary

Nov 2020 – Jun 2024

- **Vineti (Healthcare):** Developed secure healthcare platforms with React, Next.js, and Ruby under strict HIPAA compliance; improved data accuracy ~30% via observability-inspired logging and real-time sync.
- **BigBinary Academy (EdTech):** Built learning modules that lifted course completion ~30% through optimized UX flows; authored developer docs and test suites for auth, instrumentation, and API patterns.
- Improved local test reliability by introducing an HTTPS Playwright configuration adopted across multiple teams.

EDUCATION

AltCampus — Fellowship in Full-Stack Software Development (2019–2020) · MERN Stack, Ruby on Rails, Agile, Git

PROJECTS

- **Agastya Pharma** (Client) — Live corporate website for a pharmaceutical consultancy, built with Next.js (App Router), React, TypeScript, Tailwind & shadcn/ui. agastyapharma.com
- **BankEase** — Secure digital banking app integrating Plaid & Dwolla. bank-ease.vercel.app · github.com/yashjk/bank-ease
- **Apple Website Clone** — High-fidelity Apple homepage with React, GSAP, and Three.js. apple-website-gray.vercel.app · github.com/yashjk/apple_website
- **SocialSphere** — Social media app with media uploads, validation, and sharing. social-sphere-lime.vercel.app · github.com/yashjk/SocialSphere